Customer Churn Prediction Using Machine Learning

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1. Introduction

Customer churn prediction helps organizations identify customers who are likely to discontinue their services. In industries such as telecommunications and SaaS, predicting churn allows businesses to take proactive measures to retain customers and reduce revenue loss. This project develops a machine learning model capable of predicting customer churn based on customer demographics, service subscriptions, and billing information.

2. Dataset Overview

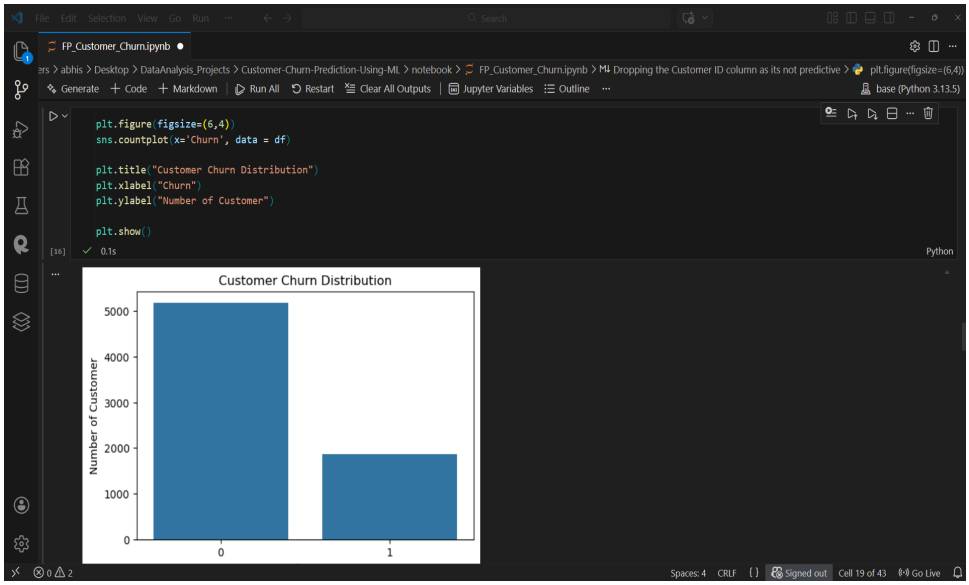
Feature	Description
customerID	Unique customer identifier
gender	Customer gender
SeniorCitizen	Whether customer is a senior citizen
Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed
MonthlyCharges	Monthly billing amount
TotalCharges	Total accumulated charges
Churn	Target variable (0 = No, 1 = Yes)

3. Data Preprocessing

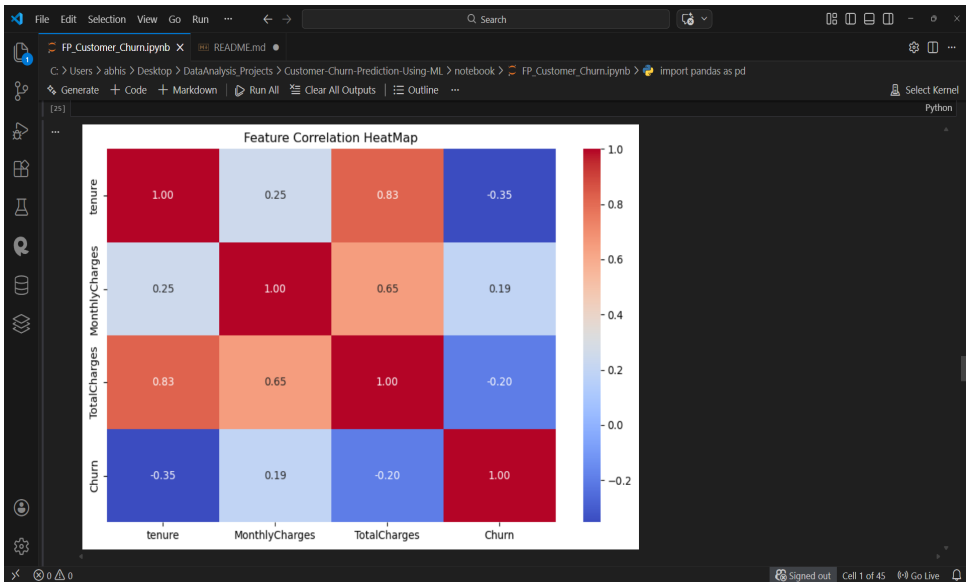
Several preprocessing steps were applied before model training. Missing values were handled, duplicate records were checked, and categorical variables were encoded into numeric form. The customerID column was removed as it does not contribute to predictive modeling.

4. Exploratory Data Analysis

Customer Churn Distribution



Feature Correlation Heatmap



Confusion Matrix

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Confusion Matrix:
 [[746 289]
 [ 81 293]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.72      0.80      1035
           1       0.50      0.78      0.61       374

    accuracy                           0.74      1409
   macro avg       0.70      0.75      0.71      1409
weighted avg       0.80      0.74      0.75      1409

ROC-AUC Score: 0.8413108062724431
```

The model achieved a recall of **0.78** for churn customers, indicating that it correctly identifies most customers who are likely to leave. Recall was prioritized because failing to identify a churn customer directly impacts business revenue. The **ROC-AUC score of 0.84** demonstrates strong class discrimination across thresholds.

5. Model Development

A Logistic Regression model was implemented for churn prediction. The dataset was split into training and testing datasets to evaluate the model's performance objectively.

6. Model Evaluation

Metric	Score
Accuracy	0.74
Precision (Churn)	0.50
Recall (Churn)	0.78
F1 Score	0.61
ROC-AUC	0.84

7. Business Insights

• Customers with longer tenure are less likely to churn. • Higher monthly charges show slightly higher churn probability. • Long-term customers contribute significantly to company revenue.

8. Conclusion

The machine learning model achieved an accuracy of 74% and a ROC-AUC score of 0.84. These results demonstrate that machine learning models can effectively identify customers who are at risk

of leaving the service. Businesses can use such predictive systems to implement proactive retention strategies.